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3.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Refer to Section 3.2

2) Horizontal IIT refers to the simultaneous export and import of the differentiated products within an industry which are of a similar quality. Vertical IIT, on the other hand, refers to the exchange of differentiated products that are of different qualities.

3) Refer to Section 3.2

Check Your Progress 2

1) Refer to Sections 3.3.1

2) "Love-of variety approach" considers that each consumer has a taste for many different varieties.

3) "Ideal variety approach", which considers that each consumer has a desire for the most preferred variety rather than many varieties.

Check Your Progress 3

1) Refer to Section 3.3.2 and Fig. 3.1

2) In the ‘integrated economy’, each variety will be produced by only one firm and therefore in only one country. All varieties will, however, be consumed in both countries, giving rise to IIT.

Check Your Progress 4

1) Refer to Section 3.3.4

2) Refer to Section 3.3.5

3) Refer to Section 3.3.3

Check Your Progress 5

1) The Grubel Lloyd (GL) index is used to measure the intensity of IIT.

2) Refer to Section 3.4.1

UNIT 4 ALTERNATIVE EXPLANATIONS OF TRADE *

Structure

4.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain new, complementary trade theories based on real world situations using concept of Economies of Scale, Imperfect Competition, Product differentiation, Technological Gap etc.;
- examine the effect of relaxing each of the classical and Heckscher-Ohlin trade theory assumptions; and
- critically examine the basics for real world international trade patterns in world market.

4.1 INTRODUCTION

The main objective of this unit is to discuss new and complementary trade theories to convey the current patterns of international trade that are closer to reality with less restrictive assumptions. Absolute Advantage, Comparative Advantage and Heckscher & Ohlin theories left unexplained a substantial share of today's international trade. Many theories have been developed after World

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War II to explain Trade and international production. These theories find out a number of factors that could explain International Trade flows between different countries, linking the aspect like cost reduction and economies of scale. Thus, this current unit explains complementary theories of international trade on the basis of economies of scale, imperfect competition and differences in technological changes between countries.

Classical trade theories explained the pattern of international trade using absolute advantage, comparative advantage and on differences in factor endowments between countries. However, these theories could not explain significant pattern of today's international trade. Relaxing the assumption of perfect competition, constant economies of scale, and no differences in technological changes between countries however required new empirical theories to provide substantial evidence of current pattern of international trade, which was not explained by classical trade theories.

The assumption of Heckscher-Ohlin theory is not generally valid as these theories assume both nations use the same technology in production. In the reality, most of the countries are using different technologies for the production. Technological gap model and product life cycle models explain these phenomena. Both of these models are viewed as dynamic extensions of the basic Heckscher-Ohlin model. These models are discussed in section 4.2.

Classical trade theories assumed constant return to scale, but international trade could be also based on increasing return to scale. According to current trade pattern in international trade, though two countries are similar in every aspect there is still a basis for mutually beneficial trade because of economies of scale. According to concept of economies of scale, when each country specializes in the production of one product, then the world total output is greater than without specialization. This was unexplained in earlier classical trade theories; (see Unit 1) hence section 4.3 will explain the concept of increasing return to scale and economies of scale to validate recent trade flows between countries.

Another significant development in recent years in international trade flows are a large portion of international trade today involves of exchange of differentiated products which the intra-industry trade model in current chapter under section 4.4 explains. Comparative advantage theory fails to explain the intra-industry trade that arise to take advantage of important economies of scale in the production in different countries.

Recently most of trade flows between countries are also affected by distance between two destinations. Gravity Model explains this concept in this chapter under section 4.5. The main focus of this model is to forecast bilateral trade flows between countries based on the distance within two countries and their respective economic dimensions. Gravity model is now used more frequently to empirically analyse trade patterns by using distance between two countries, taking the help of physical distance, cultural distance, etc.

Another important current phenomenon can be viewed from the angle of transportation and logistics costs in the international trade. Logistics cost reduce the volume and benefits of international trade. A classical theory does not consider transportation or logistics cost in the explanation of international trade. An environmental standard also affects the location of industry and international trade which is explained in this chapter at the end in section 4.7.

Krugman also significantly explained his alternative international trade theory by considering internal returns to scale and adopting a recent innovation model under monopolistic competition. Krugman's alternative theory of international trade based on more realistic assumption is explained in this chapter under section 4.6.

Relaxing the assumption of classical trade theories only modifies the concepts but does not invalidate the classical trade theories. These theories are only the extension of classical trade theories with more realistic assumptions which explain the current patterns of international trade in the world market.

4.2 TECHNOLOGICAL GAP MODEL AND PRODUCT LIFE CYCLE THEORY

According to the technological gap model introduced by Posner in 1961, most of the trade in industrialized countries is based on introducing new products and new production processes. Introduction of new product and process creates temporary monopoly for innovative firm and nation in the world market by protecting their products by patents and copyrights. The main objective of this protection is to encourage flow of inventions in the market.

According to technological gap model, technologically advanced countries export technology intensive goods. However, by technological know-how, foreigners or underdeveloped and emerging markets also adopt the foreign technology and again export their product to high technological countries with some value addition. This kind of trade occurs as labour cost is lower in the underdeveloped or in emerging markets which creates competitive advantage to emerging markets. But this model fails to explain the reasons, size of technological gaps and how technological gap is eliminated over time.

Product life cycle model explains a simplification and addition of technological gap model. The main objective of Product life of cycle theory is the improvement of trade theory beyond David Ricardo's static framework of comparative advantages. Raymond Vernon explained this theory in 1966, where he discussed three stages of production for various manufactured goods. Vernon focused on the dynamics of comparative advantage and depicted inspiration from the product life cycle to explicate how trade patterns change over time. This theory tries to explain an internationalisation process of a firm where a local manufacturer in a developed country begins selling a new, technologically advanced product to foreign markets. As demand from international markets rises, production gradually shifts overseas enabling the firm to maximise economies of scale and to avoid trade barriers, generally known as tariff jumping hypothesis[1]. In the later stage when product matures and becomes more of a commodity, the number of competitors increases, hence production shift to lower cost countries or labour-intensive countries.

He argued that comparative advantage for many products may shift over time from one country to another because these products go through a product life cycle. The main assumption of this theory is that most manufactured products follow a life cycle in which these products first appear as innovations and eventually become completely standardized. First stage of the production is called 'innovation' where the new product is introduced in to the market. In this stage, manufacturers mainly concentrate on the degree of freedom of their product and mainly invest in the product's research and Development (R &D). In the first stage the production, product is designed, developed and produced locally in the home market in limited number. Subsequently, after the establishment of the product in the home market, the foreign demand is mostly met by undertaking export.

In the second stage the production, manufactured goods became 'mature'. In the second stage of production, the price elasticity of demand for the product is comparatively low. The demand for the product increases in the foreign market and competitors appear in the international market. At this stage, producers begin manufacturing in the foreign country to supply products in the foreign market and try to decline costs with international production to compete with the competitors. Therefore, in the second stage the firm becomes international.

In the final stage product become standardized. The production technique becomes well known and reaches to the most of the competitors. Firms moved their production to lower cost and lower wage countries. The product is exported to the original country of innovation where the product is phased out in order to favour innovation of new product. Thus, the exporter becomes importer in this stage of production and national market will have to import relatively capital intensive products from low income countries.

According to product life cycle theory, when a new product is introduced in the market, its production process requisite highly skilled labour but as product matures or become standardized then production process required less skilled labour. Therefore, comparative advantage shifts from developed countries to less developed countries. The main example of product life cycle theory is the evaluation of Xerox photo-copier manufacturing plants which started originally in the USA in the 1940s and 1950s. In the first stage of production Xerox

[1]This hypothesis states that MNCs (Multinational Corporations) attempt to jump over tariff or non-tariff barriers by establishing foreign subsidiaries.

exported its products then it shifted its manufacturing plant first to Europe and finally, as the technology matured, it shifted its manufacturing plant in India in the year 1970s and 1980s.

Product life cycle theory does not apply in all conditions; mainly products with small life span may not go through the life cycle described in this theory. Secondly, this theory does not hold true in the cases of luxury goods, whose perceived value mainly depend on marketing campaigns.

4.3 ECONOMICS OF SCALE AND INTERNATIONAL TRADE

One of the significant incentives for international trade is the productivity enhancements that can arise because of economies of scale in production. After 1980's trade economists introduced the concept of economies of scale under new trade theory. One of the main assumptions of the Heckscher-Ohlin model was that products are produced under condition of constant return to scale, but the new trade theory assumes increasing return to scale, where output grows more than the increase in inputs or factors of production.

According to Economies of scale concept, the per-unit cost of producing a product falls as the scale of production rises holding input prices as constant. Therefore, larger production reflects a lower cost due to the specialization e.g., China make production at larger scale which lowers down production cost and prices of many goods. As a result of lower cost, prices of the commodities get lower and exported to other countries. Economies of scale, specialization and trade can enhance in the world productivity and welfare benefits that accrue to all trading partners.

The main assumption of this concept is under economies of scale, trade between two countries does not depend on country differences. Indeed, it is likely that countries could be equal in all respects and yet find it advantageous to trade. Thus, economies-of-scale models are frequently used to explicate trade among countries like the USA, Japan, and the European Union. Both the countries and EU and other advanced countries have similar technologies, similar endowments, and to some extent similar preferences. Expending classical models of trade, these countries would have little reason to involve in trade. However, trade between the advanced countries makes up a significant share of world trade. Economies of scale can provide an answer for this type of trade.

The idea of economies of scale can be explained by figure 4.1. Let us assume that two countries India and China are similar in every aspect and producing two commodities, wheat and cloth. In the absence of Trade India produces and consume at point A and China at point A_1. With trade, India could specialize completely in the production of commodity wheat as India has comparative advantage in the production of wheat. Similarly, China can specialize in the production of Cloth as China has comparative advantage in the production of

Cloth. By exchanging 110 quantity of wheat to 70 quantities of cloth with each other, India ends up consuming at E, and gains 20 quantities of wheat and 10 quantities of Cloth, while the China ends up consuming at E^1 and gains 30 quantities of Wheat and 10 quantities of Cloth. Thus, both the countries get extra quantities of commodities when they specialize in the production as it creates economies of scale in the production.

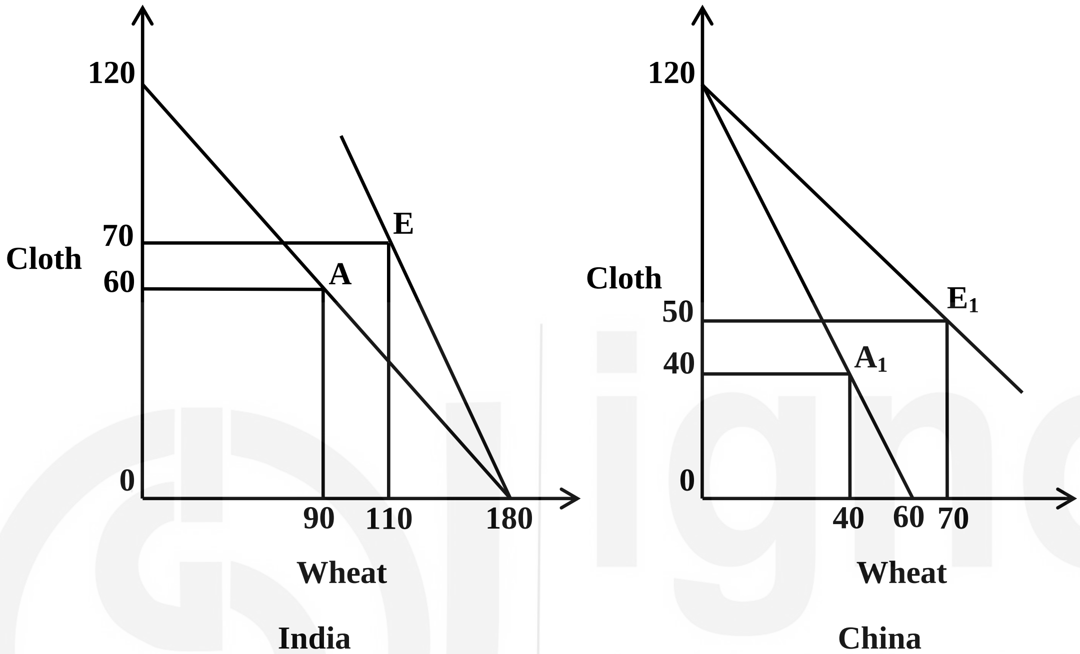

Figure 4.1: Trade based on Economies of Scale

Linder in 1961 developed an advanced hypothesis similar to economies of scale where he explained nation exports those manufactured goods for which a big domestic market exists. As there is more demand in domestic market, in the process of satisfying such a market, country obtains the necessary experience and efficiency. This experience and efficiency in manufacturing of these products further converts into increasing export of these commodities to other countries that have similar tastes, preferences and income level. The nation will import those commodities that appeal to its low and high income minorities. According to this overlapping demand hypothesis, trade in manufactures is likely to be largest among countries with similar tastes and income levels.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What is meant by technological gap model? Discuss the pattern of international trade according to technological gap model.

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